The Nodal Count Condition for the Anderson Model

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Motivation and Introduction

The Anderson Model

In 1958, P. W. Anderson (Nobel '77) introduced a toy model for electron localization/delocalization phase transition in disordered media.

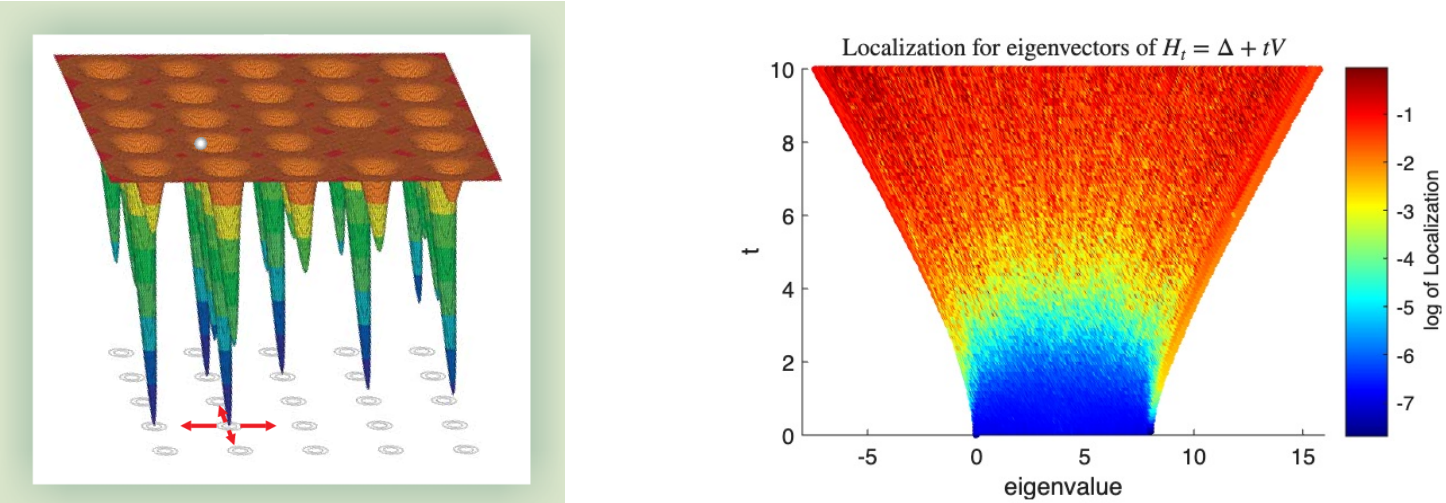


Figure: The Anderson model[1] and its phase transition[2].

[1] Picture taken from *Fifty Years of Anderson Localization*

[2] Picture taken from Aizenman-Warzel

The Anderson Model

This model is given by a random Hamiltonian, the $n \times n$ matrix

$$H_t = \Delta + tV,$$

on the graph $G = (\mathbb{Z}/L_1\mathbb{Z}) \times \cdots \times (\mathbb{Z}/L_d\mathbb{Z})$ with $n = L_1 \cdots L_d$ vertices.

- Δ is the Laplacian matrix on G.
- V is a random diagonal matrix with $V(j,j) \sim \mu$ i.i.d. for some probability distribution μ.

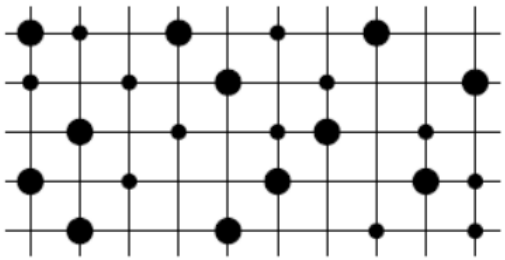


Figure: Example[3] of G for $d = 2$ with circles representing the random potential V.

[3] Picture taken from Aizenman-Warzel

Localization and Phase Transition

We distinguish two types of μ:

1. μ continuous (for example uniform on $[-1, 1]$);
2. μ discrete (for example either -1 or 1 with probability $\frac{1}{2}$).

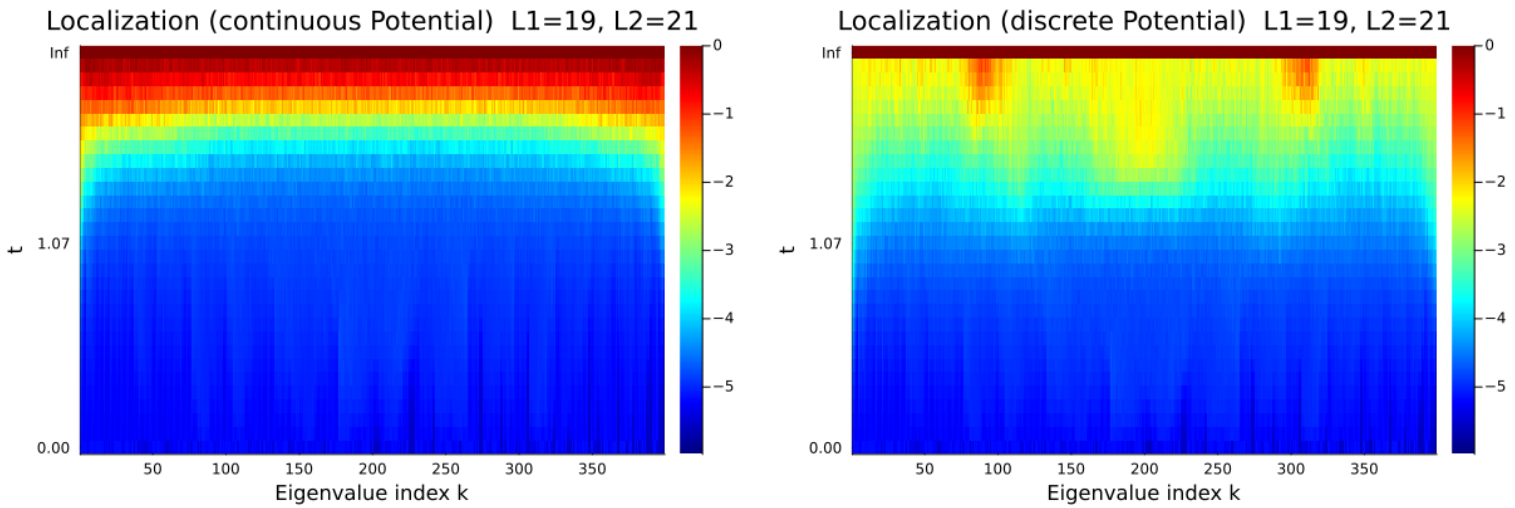


Figure: The localization phase transition for each type of distribution μ.

Known Results

So far, the problem is well understood when

1. The probability distribution μ is continuous and $d = 1, 2$;
2. The probability distribution μ is discrete and $d = 1$.

However, the tools developed so far are insufficient to tackle any other cases.

A New Tool: The Nodal Count

We investigate the differences between localized and delocalized eigenvectors through their nodal count:
The nodal count $\nu(t, k)$ is the number of sign changes of the $k-$th eigenvector of H_t.

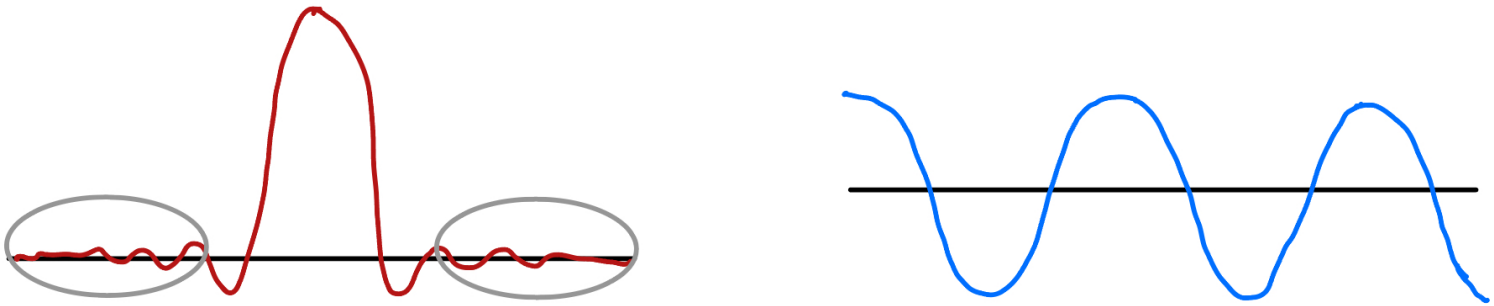


Figure: Localized eigenvectors are sensitive to noise in their tails, while delocalized eigenvectors are not.

A New Tool: The Nodal Count

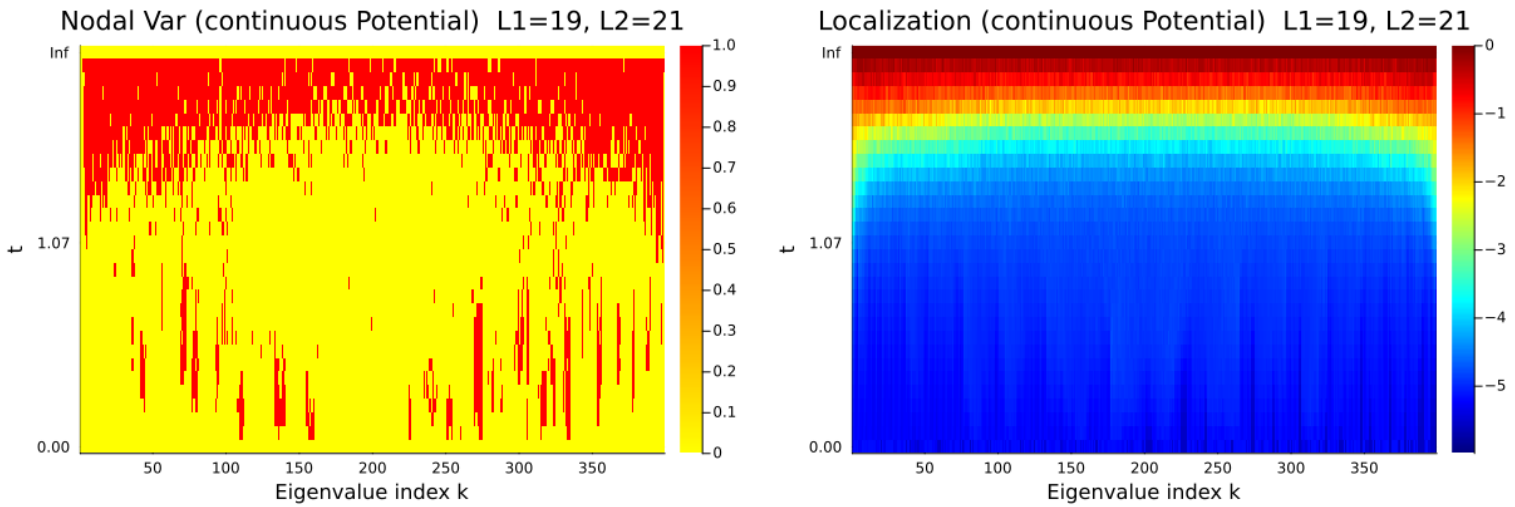


Figure: Continuous probability distribution.

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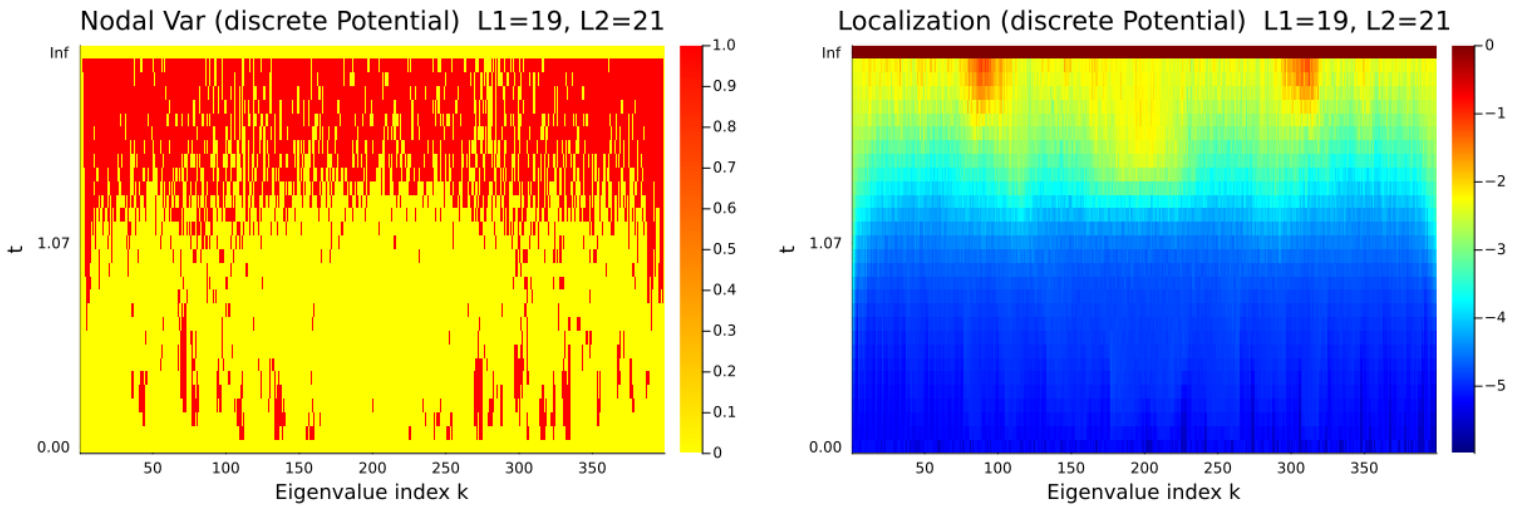


Figure: Discrete probability distribution.

Our Project: The Nodal Count Condition

Caveat: The Nodal Count is ill-defined when

- H_t has degenerate eigenvalues,
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Definition

H_t satisfies the Nodal Count Condition (NCC) if it has

- simple spectrum and
- nowhere vanishing eigenvectors.

Setup

Good and Bad Potentials

Lemma (NCC Dichotomy)

$H_t = \Delta + tV$ either fails to satisfy NCC for all $t \in \mathbb{R}$ or it satisfies NCC for all but finitely many t values.

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Definition

We call a potential V good if H_t satisfies NCC for all but finitely many t values. We call a potential bad if H_t fails to satisfy NCC for all $t \in \mathbb{R}$.

Main Question

Given the parameters:

- dimension d,
- side lengths $L_1, \ldots, L_d$,
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- side lengths $L_1, \ldots, L_d$,
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Our goal is to answer the question

Question:

What is $\mathbb{P}(V \text{ is good})$?

Nodal Count Condition for Continuous Distributions

Genericity of NCC

Theorem

If μ is a continuous distribution, then $\mathbb{P}(V \text{ is good}) = 1$ for any d and any side lengths $L_1, \ldots, L_d$.

Algebraic Independence is Good

Proposition

Suppose V is a diagonal matrix with algebraically independent diagonal entries. Then, V is good.

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- Instead of considering the perturbation $\Delta + tV$, we can analyze $V + t\Delta$.
- It suffices to find one t such that $V + t\Delta$ satisfies NCC.

Support of Eigenvectors

Let $\varphi_{k,t}$ be the $k-$th eigenvector of $V + t\Delta$.

$$\varphi_{k,t} = e_k + \sum_{j=1}^{\infty} t^j \varphi_k^{(j)}.$$

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It suffices to show that for all k and all vertices i, there exists j such that

$$\varphi_k^{(j)}(i) \neq 0.$$

Support of Eigenvectors

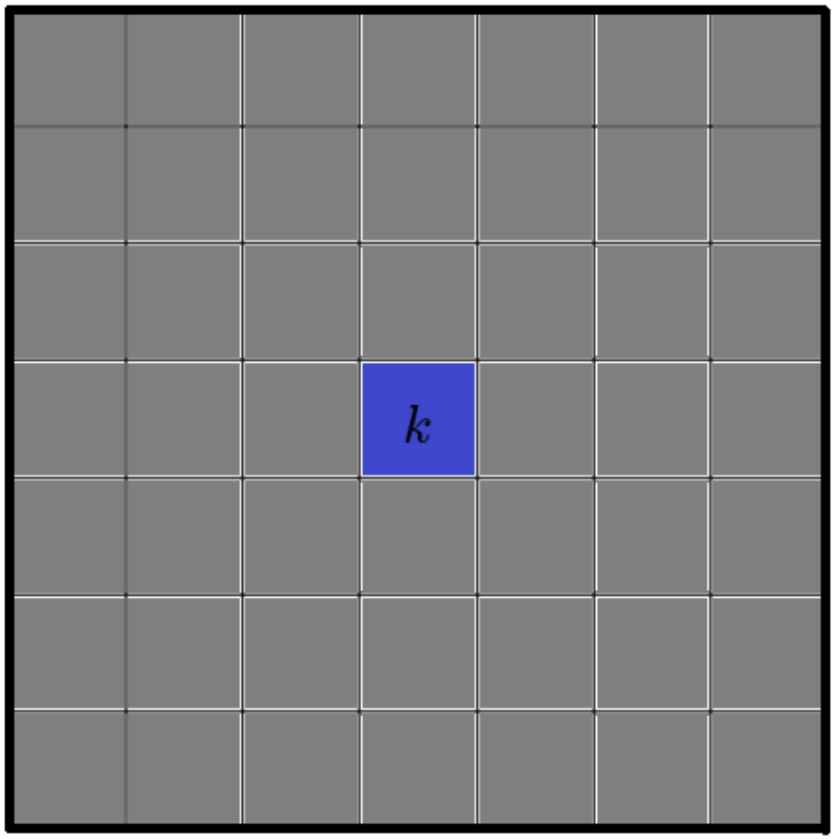


Figure: Support of $\varphi_k^{(0)}$

Support of Eigenvectors

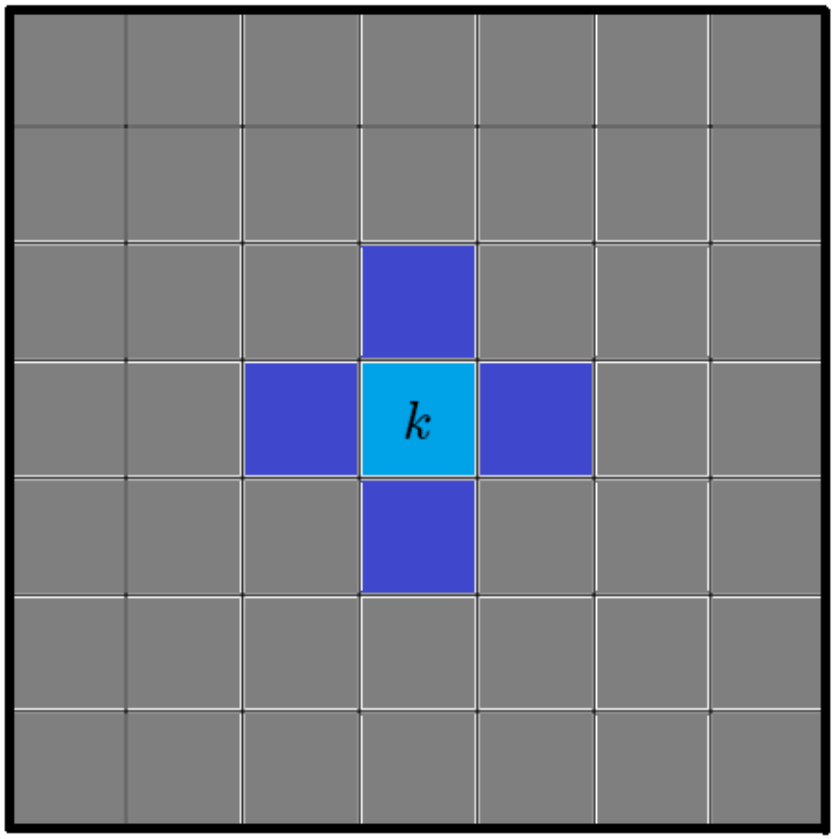


Figure: Support of $\varphi_k^{(1)}$

Support of Eigenvectors

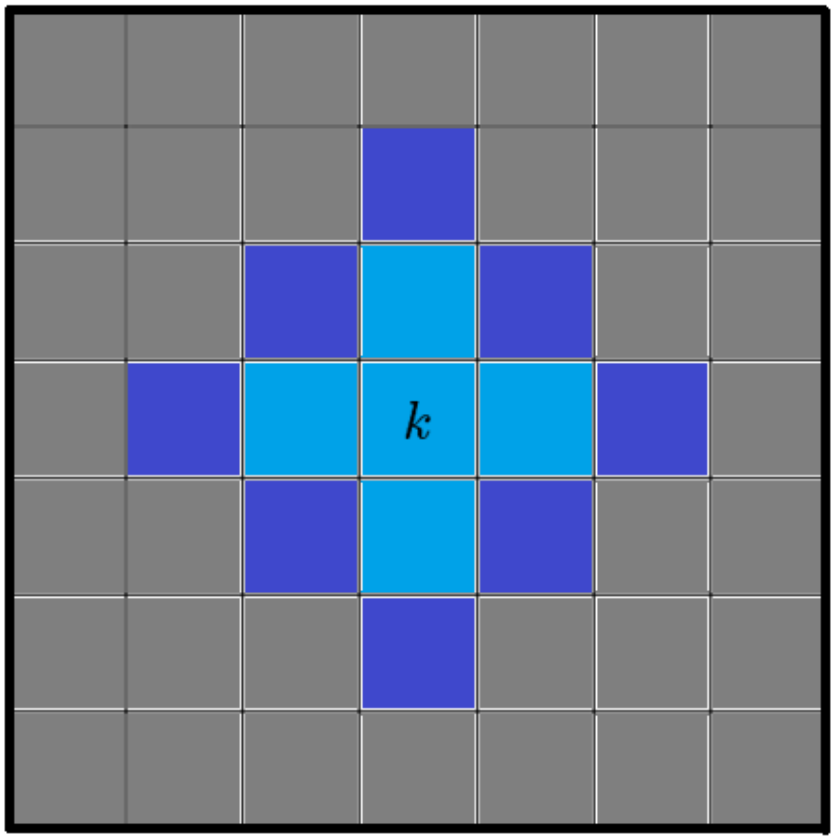


Figure: Support of $\varphi_k^{(2)}$

Support of Eigenvectors

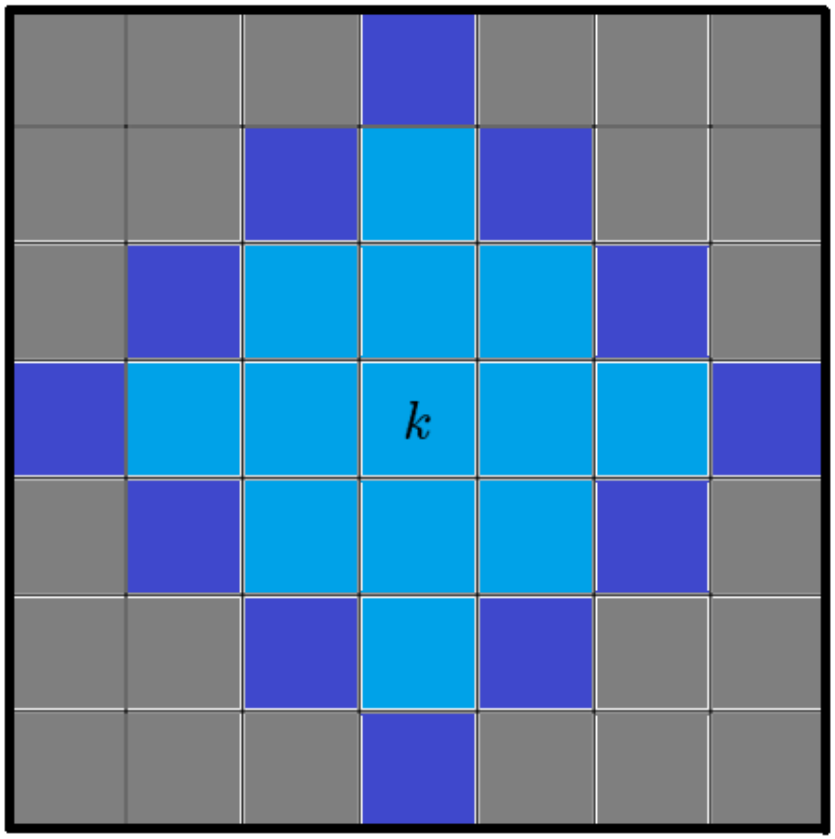


Figure: Support of $\varphi_k^{(3)}$

Support of Eigenvectors

Lemma

The $j-$th order correction for the $k-$th eigenvector at the vertex i satisfies:

$$\varphi_k^{(j)}(i) = \begin{cases} 0 & \text{if } \operatorname{dist}(i,k) > j \\ \text{nonzero} & \text{if } \operatorname{dist}(i,k) = j \end{cases}.$$

Nodal Count Condition for Discrete Distributions

Discrete Distribution

Let

$$\mu = \begin{cases} 1 & \text{with probability } \frac{1}{2} \\ -1 & \text{with probability } \frac{1}{2} \end{cases}$$

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Answer:

No! Take $V = \pm \mathrm{I}$.

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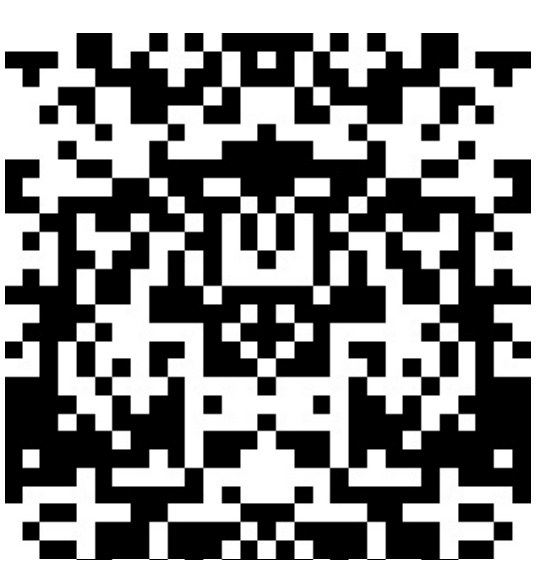
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Discrete Distribution

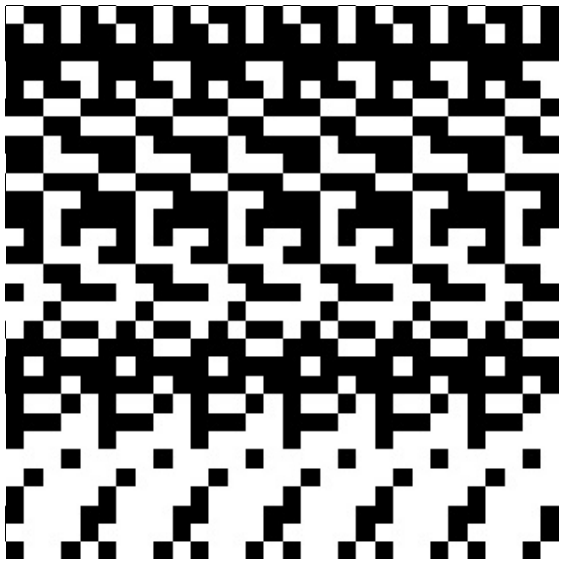
Question:

Why might $H_t(V) = \Delta + tV$ fail to satisfy the NCC for all t?

Out of the following three potentials, two are bad. Which ones?







Symmetries

Proposition

If both Δ and V commute with a nontrivial permutation matrix $\mathcal{O}$ that satisfies one of the following:

1. $\mathcal{O}^2 \neq \mathrm{I}$,
2. $\mathcal{O}$ fixes a vertex,

then V is bad.

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Note: If G has an odd number of vertices, then any shared permutation causes NCC failure for all t.

Bound on Bad Potentials

Theorem

For any d and $L_1, \ldots, L_d$ distinct odd primes, we have

$$\begin{aligned}\mathbb{P}(V \text{ is bad}) &\geq \mathbb{P}(V \text{ has shift symmetry}) \\ &= (-1)^{d-1} \cdot \sum_{J \subsetneq \{1,\ldots,d\}} (-1)^{|J|} \cdot 2^{L_J - n} > 0,\end{aligned}$$

where $L_J = \prod_{j \in J} L_j$.

$d = 1, L$ prime

Theorem

For $d = 1$ and L an odd prime, we have exactly

$$\mathbb{P}(V \text{ is bad}) = 2^{-(L-1)}(L \cdot 2^{\frac{L-1}{2}} - L + 1).$$

In particular, $\mathbb{P}(V \text{ is good}) \to 1$ as we take $L \to \infty$.

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If V has a reflection symmetry, then V is bad by the proposition.

To prove the converse, we use perturbation theory, the Discrete Fourier Transform, and algebraic facts about the $L-$th roots of unity.

$d = 1, L$ prime: Proof, cont.

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Letting

$$V_+ = \{+1 \text{ vertices}\}, \quad P_+(z) = \sum_{r \in V_+} z^r, \quad \omega = e^{\frac{2\pi k}{L}}$$

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We have

$$(\mathcal{F}^{-1} V \mathcal{F})|_{U_k} = \begin{pmatrix} P_+(1) - 1 & P_+(\omega^{2k}) \\ P_+(\omega^{-2k}) & P_+(1) - 1 \end{pmatrix}$$

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If V fails to break degeneracies, then this matrix is degenerate, so

$$P_+(\omega^{2k}) = \sum_{r \in V_+} \omega^{2kr} = 0$$

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If V breaks out degeneracies, it picks out the eigenvectors

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Taking the imaginary pary, we have

$$\sum_{r \in V_+} \left(\omega^{2k(r-j)} - \omega^{2k(j-r)} \right) = 0$$

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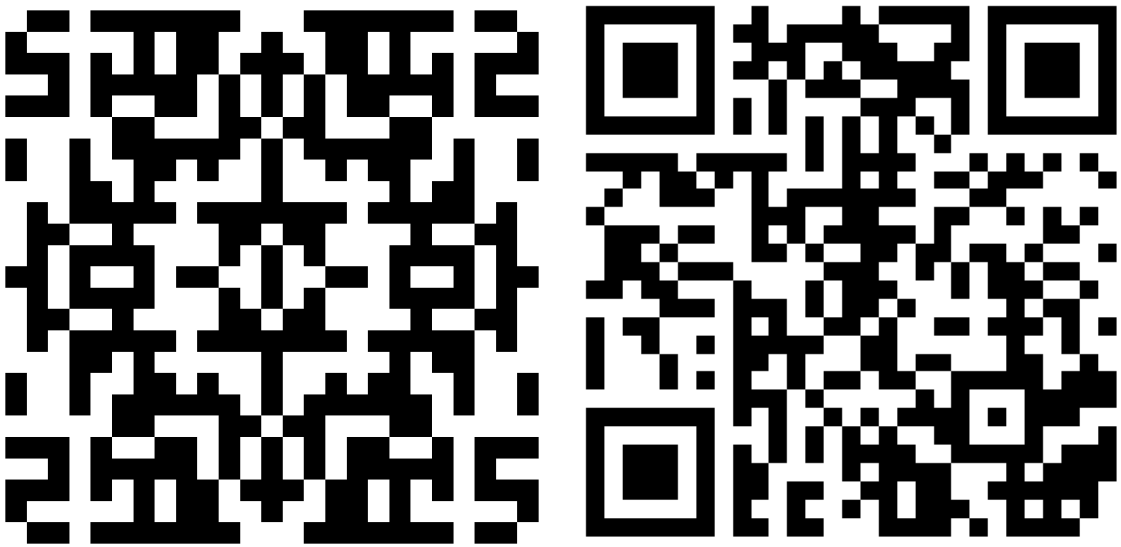
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Using this, we can show that V has reflection symmetry about the vertex j.

Find the bad potential!

Test: which potential is bad?



Thank you!

